SPORTS & ARTHROSCOPY · NEET-SS / FET / INI-SS

ACL and PCL

1. ACL anatomy and biomechanics

- **Origin:** posteromedial aspect of the **lateral femoral condyle** (medial wall of the lateral condyle, intercondylar notch). Landmarks: **lateral intercondylar ridge ("resident's ridge")** marks the anterior (upper in flexion) limit of the footprint; **lateral bifurcate ridge** separates the AM and PL bundles.
- **Insertion:** tibia, anterior to and between the intercondylar eminences, lateral to the medial tibial spine; **anterior horn of the lateral meniscus** blends with it.
- **Length about 32-38 mm, width about 10-11 mm**. **Two bundles (named by tibial insertion):**
  - **Anteromedial (AM): tight in flexion**; primary restraint to anterior tibial translation (anterior drawer at 90 degrees).
  - **Posterolateral (PL): tight in extension**; resists **rotational** loads (pivot shift) and anterior translation near extension.
- **Blood supply: middle genicular artery** (synovial sheath); **nerve:** posterior articular branches of the **tibial nerve** (mechanoreceptors: proprioception).
- **Function: primary restraint to anterior tibial translation (about 85% at 30 and 90 degrees of flexion)**; secondary restraint to internal rotation and varus/valgus.
- **Strength:** about **2,160 N** ultimate tensile load (native).

2. ACL injury

- **Mechanism: non-contact** (about 70%): **deceleration, cutting, pivoting, landing** with the knee near extension, **valgus and external (or internal) rotation**: "**pop**", immediate swelling

(**haemarthrosis: ACL tear in about 70% of acute traumatic haemarthroses** in adults).

- **Females 2-8 times higher risk: narrower notch**, **smaller ACL**, **increased Q angle/valgus**, **quadriceps dominance (less hamstring activation)**, **landing in a more upright position with knee valgus**, hormonal (preovulatory phase), **increased posterior tibial slope** (lateral). **Neuromuscular training programmes** (e.g. **FIFA 11+, PEP**) reduce ACL injury risk.
- **Examination:**
  - **Lachman test** (**most sensitive**; knee at **20-30 degrees** of flexion),
  - **Anterior drawer** (90 degrees; less sensitive acutely, hamstring guarding, posterior horn of medial meniscus acts as a "doorstop"),
  - **Pivot shift** (**most specific**; best under anaesthesia; anterior subluxation of the lateral tibial plateau in extension reducing at **20-40 degrees of flexion** via the **iliotibial band** as it moves posterior to the axis of rotation; **requires an intact IT band and MCL**),
  - **KT-1000 arthrometer** (**side-to-side difference over 3 mm** abnormal).
- **Imaging:**
  - **Radiographs: Segond fracture** (**lateral capsular avulsion of the lateral tibial plateau: anterolateral ligament (ALL)/capsule**: **pathognomonic of ACL tear**), **lateral femoral notch sign** (deep sulcus over 1.5-2 mm in the lateral femoral condyle), tibial spine avulsion (children).
  - **MRI:** discontinuity, **bone bruises ("kissing contusions") on the lateral femoral condyle (sulcus terminalis) and posterolateral tibial plateau** (pivot-shift injury), anterior tibial translation over 5-7 mm, uncovered posterior horn of the lateral meniscus, buckled PCL.
- **Associated injuries:**

- **Meniscal tears: acute: lateral meniscus more commonly torn** (posterior horn, **root tears**); **chronic ACL deficiency: medial meniscus (posterior horn) tears** increase (it becomes the secondary restraint).
- **Ramp lesions** (**meniscocapsular separation of the posterior horn of the medial meniscus**: probe through the **posteromedial portal/trans-notch view**; often missed on MRI).
- **MCL injuries** (commonest associated ligament), **posterolateral corner (PLC)** (**missed PLC = commonest cause of ACL graft failure from a missed injury**), chondral injuries, **"unhappy triad" (O'Donoghue: ACL + MCL + medial meniscus; actually lateral meniscus more often in acute injuries)**.

3. Management of ACL rupture

- **Non-operative:** "copers" (low-demand, no instability, willing to modify activities): **progressive rehabilitation**. **KANON trial (Frobell): rehabilitation + optional delayed reconstruction** gave similar 2- and 5-year outcomes to early reconstruction in young active adults (**about 50% avoided surgery**). **Increased risk of secondary meniscal tears** with recurrent instability.
- **Operative indications: recurrent instability**, **high-demand pivoting sports**, **repairable meniscal tear**, **combined ligament injuries**, **young age (children: growth-respecting techniques)**, occupational demands.
- **Timing: after resolution of swelling and recovery of full extension and quadriceps control** (avoids **arthrofibrosis**); the "**prehabilitation**" principle. **Early surgery** (within 2-3 weeks) is acceptable when ROM is good; **lock knee from a bucket-handle tear: urgent arthroscopy**.

Graft choice

Graft	Advantages	Disadvantages
Bone-patellar tendon-bone (BPTB)	**Bone-to-bone healing (6-8 weeks), lowest re-rupture in young athletes, strong (about 2,900 N)**	**Anterior knee pain, kneeling pain**, patellar fracture, patellar tendon rupture, **patella baja**, extension loss
Hamstring (semitendinosus ± gracilis, quadrupled)	**Strongest (about 4,000 N quadrupled)**, small incision, **less anterior knee pain**	**Soft-tissue healing (8-12 weeks)**, **tunnel widening**, hamstring weakness (deep flexion), saphenous nerve (infrapatellar branch) injury, **higher re-rupture in young/female athletes with small (under 8 mm) grafts**
Quadriceps tendon (± bone block)	Thick, large cross-section, **low donor morbidity**, increasing use in primary and revision	Quadriceps weakness, patellar fracture (with bone block)
Allograft	No donor morbidity, shorter surgery	**Higher failure in young (under 25), active patients** (especially **irradiated (over 2.5 Mrad)**), slower incorporation, disease transmission (rare), cost

- **Graft diameter: under 8 mm** hamstring grafts have higher revision rates.
- **Graft incorporation: avascular necrosis**, **revascularisation**, **cellular repopulation** and **remodelling (ligamentisation)**: **weakest at about 6-12 weeks**; full maturation 1-2 years.

Technical points

- **Anatomic tunnel placement**: **femoral tunnel low on the lateral wall of the notch (within the native footprint, below the lateral intercondylar ridge)**, drilled via **anteromedial portal**, **outside-in**, or flexible reamers (transtibial drilling often creates a **vertical, non-anatomic graft**: rotational instability with a persistent pivot shift).
- **Commonest technical error: femoral tunnel placed too anteriorly** (graft **tight in flexion**: loss of flexion or graft failure/stretching).
- **Tibial tunnel too anterior**: **roof impingement in extension** (loss of extension, **cyclops lesion** risk); too posterior: vertical graft, PCL impingement.
- **Fixation:** interference screws, suspensory buttons (femoral), staples/screw-post (tibial).
- **Lateral extra-articular tenodesis (LET; modified Lemaire)** or **ALL reconstruction** added in **high-risk patients** (young, high-grade pivot shift, return to pivoting sport, generalised laxity, revision): **STABILITY trial** (hamstring ACLR + LET in young patients: **reduced graft rupture/persistent rotatory laxity**).
- **Double-bundle reconstruction:** no clear clinical advantage over anatomic single-bundle.
- **ACL repair** (proximal avulsion types) is being revisited with augmentation; not standard.

Rehabilitation and return to sport

- **Early full extension, quadriceps activation, closed kinetic chain exercises** (less ACL graft strain than open chain in the early phase); **avoid open-chain quadriceps (knee extension) exercises between 0-45 degrees** early.
- **Return to sport: criteria-based** (strength over 90% limb symmetry index, hop tests, psychological readiness) and **not before 9 months**: each month delay in return to sport until 9 months reduces re-injury by about 51% (Grindem, Delaware-Oslo).

- **Contralateral ACL injury** risk is similar to graft rupture risk in young athletes.

Complications

- **Arthrofibrosis/loss of extension** (commonest complication; **cyclops lesion** = fibroproliferative nodule anterior to the graft: **loss of terminal extension with a "clunk"**: arthroscopic excision).
- **Graft failure: early (technical: tunnel malposition, fixation failure)**, **late (re-injury)**; **missed PLC/medial injuries**; **increased posterior tibial slope (over 12 degrees)**.
- **Infection** (**coagulase-negative staphylococci**, *S. aureus*; **pre-soak graft in vancomycin** reduces infection), **tunnel widening**, DVT, saphenous nerve injury, anterior knee pain, **osteoarthritis** (higher in meniscectomy cases; reconstruction does not prevent OA).
- **Revision ACLR:** assess **tunnel position, widening (over 15 mm: staged bone grafting)**, **posterior tibial slope (over 12 degrees: slope-reducing osteotomy)**, **varus alignment (HTO)**, missed associated laxities.

4. Paediatric ACL injuries

- **Tibial spine (eminence) fractures** (**Meyers and McKeever classification**: I undisplaced, II hinged anterior elevation, III completely displaced (IIIA, IIIB rotated), IV comminuted): **types I and reducible II = closed reduction in extension, cast**; **III/IV and irreducible II = arthroscopic/open fixation (sutures or screws)**; **entrapment of the intermeniscal ligament or anterior horn of medial meniscus** blocks reduction.
- **Intrasubstance ACL tears in skeletally immature:** early reconstruction (to avoid meniscal/chondral damage) using **physeal-sparing (all-epiphyseal, Micheli-Kocher extraphyseal ITB)** or **transphyseal** techniques (with soft-tissue grafts, small vertical central tunnels, no hardware across the physis) depending on

remaining growth (Tanner stage, bone age); risk of **growth disturbance (angular deformity, leg-length discrepancy)**.

5. PCL anatomy and biomechanics

- **Origin: lateral surface of the medial femoral condyle** (anterior notch); **insertion: posterior tibia in a central depression (PCL facet), about 1-1.5 cm below the joint line**.
- **Twice as strong and wider than the ACL** (about 13 mm wide); **two bundles (named by femoral origin): anterolateral (AL, larger, tight in flexion)** and **posteromedial (PM, tight in extension)**.
- **Meniscofemoral ligaments** (from the posterior horn of the lateral meniscus to the medial femoral condyle): **Humphrey (anterior to the PCL)** and **Wrisberg (posterior to the PCL)** (at least one in about 90% of knees).
- **Primary restraint to posterior tibial translation (about 90-95%)**; secondary restraint to external rotation (with PLC).

6. PCL injury

- **Mechanism: dashboard injury** (posteriorly directed force on the proximal tibia with the knee flexed), **fall onto a flexed knee with the foot plantarflexed**, **hyperextension** (combined injuries).
- **Examination:**
  - **Posterior drawer test** (**most sensitive**; 90 degrees of flexion): **grading by tibial step-off** relative to the femoral condyles (**normal about 1 cm anterior step-off**): **grade I: 0-5 mm translation (tibia still anterior to condyles)**, **grade II: 5-10 mm (tibia flush with the condyles)**, **grade III: over 10 mm (tibia posterior to the condyles: suggests combined PCL + PLC/posteromedial injury)**.
  - **Posterior sag (Godfrey) sign**, **quadriceps active test** (contraction reduces posterior subluxation), **reverse pivot shift** (PLC).

  - **Dial test** (external rotation asymmetry over 10 degrees): **at 30 degrees only = isolated PLC**, **at 30 and 90 degrees = PLC + PCL**.
- **Imaging: stress radiographs** (posterior displacement **over 12 mm** side-to-side suggests combined injury), MRI, **avulsion fractures of the tibial PCL insertion**.

Management

- **Isolated grade I-II PCL injuries: non-operative**: **quadriceps-strengthening rehabilitation**, **PCL brace (dynamic anterior drawer force, e.g. Jack brace)**: good functional outcomes; natural history may include **medial compartment and patellofemoral OA** in the long term.
- **Bony avulsions of the PCL tibial insertion: early fixation** (posterior approach (Burks and Schaffer) or arthroscopic) gives excellent results.
- **Grade III, combined injuries (PLC/multiligament), symptomatic chronic instability: reconstruction**: **transtibial** (**"killer turn"** of the graft at the posterior tibia: graft abrasion/attenuation) vs **tibial inlay** (open posterior bone block; avoids the killer turn); **single- (AL bundle)** or **double-bundle**.
- **Risks: popliteal artery injury** during tibial tunnel drilling (**guide-wire penetration; keep the knee flexed at 90 degrees**, use a guard, posteromedial portal to visualise).

7. Posterolateral corner (PLC)

- **Components: lateral (fibular) collateral ligament (LCL)**, **popliteus tendon**, **popliteofibular ligament** (the "big three"), plus lateral capsule, arcuate ligament, biceps femoris, IT band, lateral gastrocnemius.
- **Function:** primary restraint to **varus** and **external rotation** (especially at 30 degrees).

- **Injury:** posterolateral force to the anteromedial tibia, hyperextension/varus; **common peroneal nerve injury in about 15-30%**.
- **Signs: varus thrust gait**, **dial test**, **external rotation recurvatum test**, **posterolateral drawer**, **reverse pivot shift**, **varus stress at 30 degrees (and at 0 degrees: suggests combined cruciate injury)**.
- **Imaging: arcuate sign** (**avulsion of the fibular styloid**: attachments of the popliteofibular, arcuate and fabellofibular ligaments): PLC injury; varus stress radiographs.
- **Treatment: acute repair within about 2-3 weeks** (higher failure of repair alone) or **reconstruction (anatomic, e.g. LaPrade)**: **reconstruction more reliable than repair**; address **varus alignment (HTO first)** in chronic cases; **PLC must be addressed with any cruciate reconstruction** (unrecognised PLC injury causes cruciate graft failure).

8. Multiligament knee injury/knee dislocation

- **Schenck classification: KD I** single cruciate + collateral; **KD II** both cruciates; **KD III M/L** both cruciates + **MCL (M) or PLC (L)**; **KD IV** both cruciates + both collaterals; **KD V** fracture-dislocation; C = arterial injury, N = nerve injury.
- **Vascular injury (popliteal artery)** in about **10-30%** (anterior dislocations: intimal stretch; posterior: transection). **ABI under 0.9: CT angiography/vascular surgery**; **normal pulses do not exclude injury**: serial **ankle-brachial index (ABI)** ± angiography. Ischaemia over 6-8 hours: high amputation rate; **prophylactic fasciotomy** after revascularisation.
- **Peroneal nerve injury** in about 25% (especially **KD III L**).
- **Management:** reduce urgently, assess vascular status, **external fixation** for unstable/vascular/open injuries; **early (within 3 weeks) single-stage or staged ligament repair/reconstruction** generally

gives better results than non-operative treatment; **arthrofibrosis** is common.

Exam pearl: **Lachman most sensitive; pivot shift most specific (needs intact ITB/MCL). Segond fracture = ACL tear. Acute ACL: lateral meniscus tears; chronic: medial meniscus. Commonest technical error: femoral tunnel too anterior (tight in flexion). Tibial tunnel too anterior: roof impingement, cyclops lesion.**

Exam pearl: **AM bundle tight in flexion, PL bundle tight in extension (ACL named by tibial insertion); PCL AL bundle tight in flexion (named by femoral origin). Humphrey anterior, Wrisberg posterior to the PCL.**

Exam pearl: **PCL grade III (tibia behind condyles) = suspect PLC. Dial test: 30 only = PLC; 30 and 90 = PLC + PCL. Arcuate sign = fibular styloid avulsion. Missed PLC causes cruciate graft failure. Knee dislocation: check ABI; under 0.9 = angiography.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.